



Microsoft Power BI

Complete Study Notes & Full Explanation Guide

BUSINESS INTELLIGENCE · DATA MODELING · DAX · VISUALIZATION

Table of Contents

1. Introduction to Power BI	03
2. Power BI Components / Ecosystem	03
3. Power BI Workflow	04
4. Data Sources	04
5. Power Query (Get & Transform)	05
6. Data Modeling	06
7. DAX – Data Analysis Expressions	07
8. Visualizations	09
9. Reports vs Dashboards	10
10. Filters & Slicers	10
11. Power BI Service (Cloud)	11
12. Row-Level Security (RLS)	12
13. Power BI Mobile & Gateway	12
14. Best Practices & Performance Tips	13
15. Common Interview / Exam Q&A	13

1. Introduction to Power BI

Power BI is a business intelligence (BI) and data visualization tool developed by Microsoft. It allows users to connect to hundreds of data sources, clean and transform raw data, build data models, and create interactive reports and dashboards that turn data into actionable insights. It is widely used by data analysts, business analysts, and decision-makers to monitor KPIs (Key Performance Indicators), track business performance, and make data-driven decisions without needing deep programming knowledge.

Why Power BI?

- **Easy to learn:** Drag-and-drop interface, similar to Excel, so business users can pick it up quickly.
 - **Wide connectivity:** Connects to Excel, SQL Server, Azure, SharePoint, Google Analytics, web APIs, and hundreds of other sources.
 - **Powerful data modeling:** Built-in engine (VertiPaq) that compresses and stores millions of rows efficiently in memory.
 - **Rich visuals:** Native and custom visuals (charts, maps, KPI cards, matrices) for storytelling with data.
 - **Cost-effective:** Free desktop application; affordable licensing for cloud sharing (Power BI Pro / Premium).
 - **Integration:** Works seamlessly with the Microsoft ecosystem — Excel, Teams, SharePoint, Azure, and Office 365.
- Key Idea:** Power BI follows the philosophy of "*Connect → Transform → Model → Visualize → Share*". This single flow is the backbone of every Power BI project.

2. Power BI Components / Ecosystem

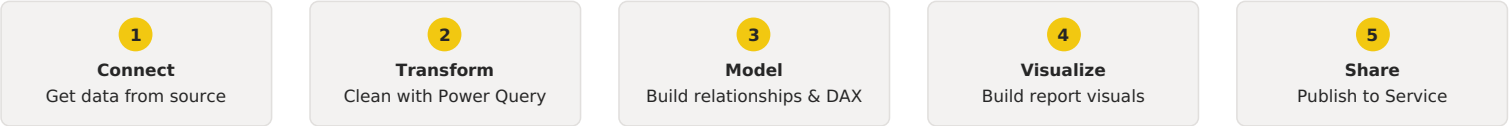
Power BI is not a single application — it is a suite of interconnected tools, each serving a different purpose in the BI lifecycle.

Component	Description
Power BI Desktop	A free Windows application used to connect to data, transform it (Power Query), build data models, write DAX measures, and design reports. This is where 80% of development work happens.
Power BI Service	A cloud-based SaaS platform (app.powerbi.com) used to publish, share, and collaborate on reports and dashboards. Includes workspaces, apps, and scheduled data refresh.
Power BI Mobile	Mobile apps (iOS/Android/Windows) that let users view reports and dashboards, and receive data alerts on the go.

Power BI Report Server	An on-premises server for organizations that cannot use the cloud service due to compliance or security policies.
Power BI Gateway	A bridge software that securely connects on-premises data sources (like local SQL Server) to the Power BI cloud service for scheduled refresh.
Power BI Embedded	Azure service allowing developers to embed Power BI reports directly into custom applications.
Power BI Report Builder	A tool for creating paginated (pixel-perfect) reports, ideal for printable invoices and operational reports.

3. Power BI Workflow

Every Power BI project generally follows the same five-stage pipeline:



1. **Connect:** Import or connect live to a data source (Excel, SQL, web, cloud services).
2. **Transform:** Use Power Query Editor to clean, reshape, filter, merge, and append data.
3. **Model:** Define relationships between tables and create calculated columns/measures using DAX.
4. **Visualize:** Drag fields onto the canvas to build charts, tables, and KPI visuals.
5. **Share:** Publish the .pbix file to the Power BI Service, and share via dashboards, apps, or embedded links.

4. Data Sources

Power BI can connect to over 200+ data sources. These are broadly categorized as follows:

Category	Examples
Files	Excel, CSV, XML, JSON, PDF, Text, Folder
Databases	SQL Server, Oracle, MySQL, PostgreSQL, IBM DB2, Teradata
Azure	Azure SQL Database, Azure Synapse, Azure Data Lake, Azure Blob Storage
Online Services	SharePoint Online, Dynamics 365, Salesforce, Google Analytics, GitHub
Other	Web (API/HTML), OData Feed, ODBC, R/Python scripts, Blank Query

Connectivity Modes

- **Import Mode:** Data is copied and compressed into Power BI's internal in-memory engine (VertiPaq). Fastest performance, but requires manual/scheduled refresh.
- **DirectQuery:** Power BI sends live queries to the source database instead of storing data locally. Always up to date, but performance depends on the source system.
- **Live Connection:** Similar to DirectQuery but specifically used with Analysis Services / Power BI datasets — only visuals are built, the model itself lives elsewhere.
- **Composite Model:** Combines Import and DirectQuery tables within a single data model.

Note: Import mode is recommended for most use cases due to superior query performance, since VertiPaq's columnar compression can handle millions of rows in memory.

5. Power Query (Get & Transform)

Power Query is Power BI's data preparation (ETL – Extract, Transform, Load) engine. It uses a functional language called **M language** behind the scenes, though most transformations can be done using the graphical Power Query Editor without writing any code.

Common Transformations

- **Remove/Keep Rows or Columns:** Filter unnecessary data.
- **Change Data Type:** Convert text to number, date, etc.
- **Split/Merge Columns:** Split a "Full Name" column into First and Last Name.
- **Group By:** Aggregate data (sum, average, count) by category.
- **Pivot/Unpivot Columns:** Reshape data between wide and long formats — unpivoting is very common for preparing data for reporting.

- **Merge Queries:** Similar to SQL JOIN — combine two tables based on a common column.
- **Append Queries:** Similar to SQL UNION — stack rows of two tables with the same structure.
- **Conditional Columns:** Create new columns using IF-based logic through the UI.
- **Custom Columns:** Add computed columns using M formulas.

The Applied Steps Panel

Every transformation performed in Power Query is recorded as a step in the **Applied Steps** panel. This creates a repeatable, auditable script (written in M) that re-executes automatically every time the data is refreshed — meaning you clean the data once, and Power BI repeats the process forever.

Tip: Always rename your queries and steps clearly. A clean Applied Steps list makes debugging and maintaining reports significantly easier later on.

6. Data Modeling

Data modeling is the process of defining how tables relate to each other so Power BI can correctly aggregate and filter data across the entire report.

Star Schema (Recommended Design)

The best-practice data model in Power BI is the **Star Schema**, which consists of:

- **Fact Tables:** Contain transactional/numeric data (e.g., Sales, Orders) — typically large tables with foreign keys and measures.
- **Dimension Tables:** Contain descriptive attributes (e.g., Customer, Product, Date, Region) used to slice and filter the fact table.

In a star schema, dimension tables surround a central fact table, connected by one-to-many relationships — visually resembling a star shape. This design is favored over a "snowflake" (normalized) schema because it is simpler and performs faster in Power BI's engine.

Relationships

Cardinality	Meaning
One-to-Many (1:*)	Most common; one row in dimension table relates to many rows in fact table (e.g., one Product → many Sales).
One-to-One (1:1)	Each row in Table A matches exactly one row in Table B.
Many-to-Many (*:*)	Used carefully; occurs when neither table has unique keys, handled via bridge tables ideally.

Cross-Filter Direction

- **Single:** Filters flow one way (dimension → fact). Recommended default for performance and predictability.
- **Both:** Filters flow in both directions. Useful in specific scenarios but can cause ambiguity and slower performance.

The Date Table

A dedicated **Date table** is essential for any time-intelligence analysis (e.g., Year-over-Year growth). It can be created manually, via Power Query, or using the DAX function `CALENDAR ()` or `CALENDARAUTO ()` , and should be marked as the official "Date Table" in model settings.

7. DAX - Data Analysis Expressions

DAX is the formula language used in Power BI to create calculated columns, measures, and custom tables. It is similar in syntax to Excel formulas but is designed to work with relational data models rather than single cells.

Calculated Columns vs Measures

Aspect	Calculated Column	Measure
Computed	Row-by-row, at data refresh	On-the-fly, at query/visual time
Storage	Stored in the model (uses memory)	Not stored; calculated dynamically
Context	Row context	Filter context
Use Case	Categorization, static grouping	Aggregations, KPIs, dynamic calculations

Best Practice: Prefer measures over calculated columns wherever possible — they keep the model lighter and more performant since they aren't stored in memory.

Row Context vs Filter Context

- **Row Context:** Exists when a formula evaluates row-by-row, such as in a calculated column, or inside iterator functions like `SUMX`.
- **Filter Context:** The set of filters applied to the data before a calculation happens — determined by slicers, rows/columns in a visual, and filter cards.

Common DAX Functions

Function	Category	Purpose
<code>SUM()</code>	Aggregation	Adds all numbers in a column.
<code>AVERAGE()</code>	Aggregation	Returns the arithmetic mean.
<code>COUNT()</code> / <code>COUNTA()</code>	Aggregation	Counts numeric / non-blank values.
<code>CALCULATE()</code>	Filter modifier	The most powerful DAX function — evaluates an expression under a modified filter context.
<code>FILTER()</code>	Filter	Returns a filtered table for use inside other functions.
<code>ALL()</code>	Filter modifier	Removes filters from a table or column, useful for "% of total" calculations.
<code>SUMX()</code> / <code>AVERAGEX()</code>	Iterator	Evaluates an expression row-by-row, then aggregates the result.
<code>RELATED()</code>	Relationship	Pulls a value from a related table (used in calculated columns).
<code>IF()</code>	Logical	Conditional logic, same as Excel.
<code>SWITCH()</code>	Logical	Cleaner alternative to nested IFs.
<code>DATEADD()</code> / <code>SAMEPERIODLASTYEAR()</code>	Time Intelligence	Shifts dates to compare across periods (e.g., YoY analysis).
<code>TOTALYTD()</code> / <code>TOTALQTD()</code> / <code>TOTALMTD()</code>	Time Intelligence	Calculates running totals for Year/Quarter/Month-to-date.
<code>DIVIDE()</code>	Math	Safe division that automatically handles divide-by-zero errors.

Example DAX Measures

```
Total Sales = SUM(Sales[SalesAmount])

Total Profit = SUM(Sales[SalesAmount]) - SUM(Sales[CostAmount])

YoY Growth % =
DIVIDE(
    [Total Sales] - CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date])),
    CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
)

Running Total =
CALCULATE(
    [Total Sales],
    FILTER(ALLSELECTED('Date'), 'Date'[Date] <= MAX('Date'[Date]))
)
```

8. Visualizations

Power BI offers a wide variety of built-in visuals, and thousands more can be imported from the AppSource marketplace as custom visuals.

Visual Type	Best Used For
Bar / Column Chart	Comparing values across categories.
Line Chart	Showing trends over time.
Pie / Donut Chart	Showing proportion of a whole (best with few categories).
Card / Multi-row Card	Highlighting a single key metric or KPI.
Table / Matrix	Displaying detailed, tabular data with subtotals (Matrix supports pivoting).
Map / Filled Map	Visualizing geographic data.
Gauge	Showing progress toward a target/goal.
Scatter Chart	Showing correlation between two numeric variables.
Waterfall Chart	Showing sequential positive/negative contributions to a total.
Treemap	Showing hierarchical, proportional data in nested rectangles.
KPI Visual	Comparing a value against a target with a trend indicator.

Conditional Formatting & Drill-Through

- **Conditional Formatting:** Apply color scales, data bars, or icons to table/matrix cells based on values.
- **Drill Down/Up:** Navigate a visual's hierarchy (e.g., Year → Quarter → Month) using the arrows in the visual header.
- **Drill-through:** Right-click a data point to jump to a separate, detailed report page filtered to that context.
- **Tooltips:** Custom report pages that appear when hovering over a data point, showing extra detail.

9. Reports vs Dashboards

Aspect	Report	Dashboard
Pages	Multiple pages/tabs	Single page (canvas of tiles)
Interactivity	Fully interactive (filters, slicers, drill-through)	Mostly static tiles pinned from reports
Created In	Power BI Desktop or Service	Power BI Service only
Source	A single dataset	Can combine tiles from multiple reports/datasets
Purpose	Deep analysis & exploration	High-level, at-a-glance monitoring

Note: A Dashboard is built by "pinning" visuals from one or more Reports. Reports are the detailed source; Dashboards are the executive summary view.

10. Filters & Slicers

Power BI applies filters in a specific hierarchy, from broadest to narrowest scope:

1. **Report-level filters:** Apply to every page in the report.
2. **Page-level filters:** Apply only to the current page.
3. **Visual-level filters:** Apply only to a single visual.
4. **Drill-through filters:** Applied automatically when navigating from a source visual.

Slicers

Slicers are on-canvas visual filters that let end users interactively filter the report (e.g., a dropdown or list of Region names). Unlike the Filters pane, slicers are always visible to the report viewer.

Sync Slicers

The "Sync Slicers" feature allows a single slicer selection to apply across multiple report pages, keeping filters consistent as users navigate.

11. Power BI Service (Cloud)

The **Power BI Service** (app.powerbi.com) is where reports are published, shared, and consumed by an organization.

Key Concepts

- **Workspace:** A collaborative container where teams develop and store related dashboards, reports, and datasets before publishing to consumers.
- **App:** A polished, packaged, read-only collection of dashboards and reports distributed to a broad audience within the organization.
- **Dataset:** The underlying data model (published from a .pbix file) that reports are built on; can be reused across multiple reports.
- **Scheduled Refresh:** Automatically updates Import-mode datasets on a defined schedule (up to 8 times/day on Pro, 48 times/day on Premium).
- **Q&A:** A natural-language query feature — users type a question in plain English and Power BI auto-generates a visual.
- **Data Alerts:** Notifications triggered when a KPI/card visual crosses a defined threshold.
- **Subscriptions:** Emailed snapshots of a report/dashboard sent on a recurring schedule.

Licensing Tiers

License	Highlights
Free	Personal use, no sharing capability with other Pro-less users.
Pro	Per-user license enabling sharing, collaboration, and app distribution.
Premium (Per Capacity / Per User)	Dedicated cloud capacity, larger datasets, more frequent refresh, paginated reports, and AI features.

12. Row-Level Security (RLS)

Row-Level Security restricts data access for given users at the row level within a shared dataset, so that different users see only the data relevant to them (e.g., a regional manager sees only their region's sales).

How It Works

1. Define **Roles** in Power BI Desktop (Modeling → Manage Roles).
2. Write a DAX filter expression for each role, e.g. `[Region] = "North"` .
3. Publish the report to the Service.
4. Assign specific users or Azure AD security groups to each role in the workspace settings.

Tip: Use `USERNAME()` or `USERPRINCIPALNAME()` combined with a mapping table for dynamic RLS, so a single role can serve many different users automatically.

13. Power BI Mobile & Gateway

Power BI Mobile

Available for iOS, Android, and Windows. Allows users to view dashboards/reports, annotate and share snapshots, and receive push notifications for data alerts — ideal for executives who need on-the-go KPI visibility.

Power BI Gateway

Two types of gateways bridge on-premises data with the cloud service:

- **Personal Gateway:** For individual use; only the installing user can manage refresh schedules; supports Import mode only.
- **Enterprise (Standard) Gateway:** Shared by multiple users/reports across an organization; supports both Import and DirectQuery, and is managed centrally by IT.

14. Best Practices & Performance Tips

- **Use Star Schema:** Avoid wide, flat, denormalized tables; separate facts and dimensions.
- **Reduce Cardinality:** High-cardinality columns (like unique IDs) consume more memory — remove unnecessary unique columns.
- **Prefer Measures over Calculated Columns:** Keeps the data model smaller and faster.
- **Disable Auto Date/Time:** Turn off Power BI's automatic hidden date tables for large models; use one shared Date table instead.
- **Filter Data Early:** Remove unused columns and rows in Power Query before loading into the model, not after.
- **Use Variables in DAX (`VAR` / `RETURN`):** Improves readability and can improve performance by avoiding repeated calculations.
- **Limit Visuals per Page:** Too many visuals on one page slow down rendering; split complex analysis across multiple pages.
- **Use Bookmarks & Buttons:** For building intuitive navigation and interactive storytelling experiences.
- **Rename & Organize Fields:** Use display folders and clear naming conventions so report consumers can self-serve easily.

15. Common Interview / Exam Q&A

Q1: What is the difference between Power BI Desktop and Power BI Service?

Power BI Desktop is the free authoring tool used to build reports and models locally. Power BI Service is the cloud platform used to publish, share, collaborate, and schedule refresh for those reports.

Q2: What is VertiPaq?

VertiPaq is Power BI's in-memory columnar storage and compression engine that powers Import mode, allowing extremely fast aggregation over large datasets.

Q3: What's the difference between CALCULATE and FILTER?

`CALCULATE` modifies the filter context of a measure and is the primary function for building complex logic; `FILTER` returns a filtered table, typically used as an argument inside functions like `CALCULATE` or `SUMX` .

Q4: What is a slicer, and how is it different from a filter pane?

A slicer is a visual, on-canvas filtering control visible to report consumers, while the Filters pane is a background panel used by the report author to apply filters that may or may not be visible to consumers.

Q5: Why is Import mode generally faster than DirectQuery?

Import mode loads a compressed copy of the data into Power BI's in-memory VertiPaq engine, so queries run against local memory. DirectQuery sends a live query to the source system for every interaction, so speed depends entirely on the performance of the source database.

Q6: What is the purpose of a Date/Calendar table?

It provides a single, continuous, and reliable source of dates for time-intelligence functions (YTD, MTD, YoY comparisons) and ensures consistent date-based filtering across all fact tables in the model.